(FP2) 4. Integration 专题作业2

NO. 1

OR

The polar equation of a curve C is $r = a(1 + \cos\theta)$ for $0 \leq \theta < 2\pi$, where a is a positive constant.

i. Sketch C. [2]

ii. Show that the cartesian equation of C is [2]

$$x^2 + y^2 = a\left(x + \sqrt{(x^2 + y^2)}\right).$$

.....

iii. Find the area of the sector of C between $\theta = 0$ and $\theta = \frac{1}{3}\pi$. [4]

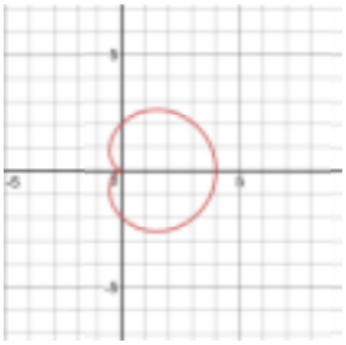
.....

iv. Find the arc length of C between the point where $\theta = 0$ and the point where $\theta = \frac{1}{3}\pi$. [5]

.....

A level FURTHER MATHEMATICS 2017/Winter/Paper1_1/Q11O

Mark scheme

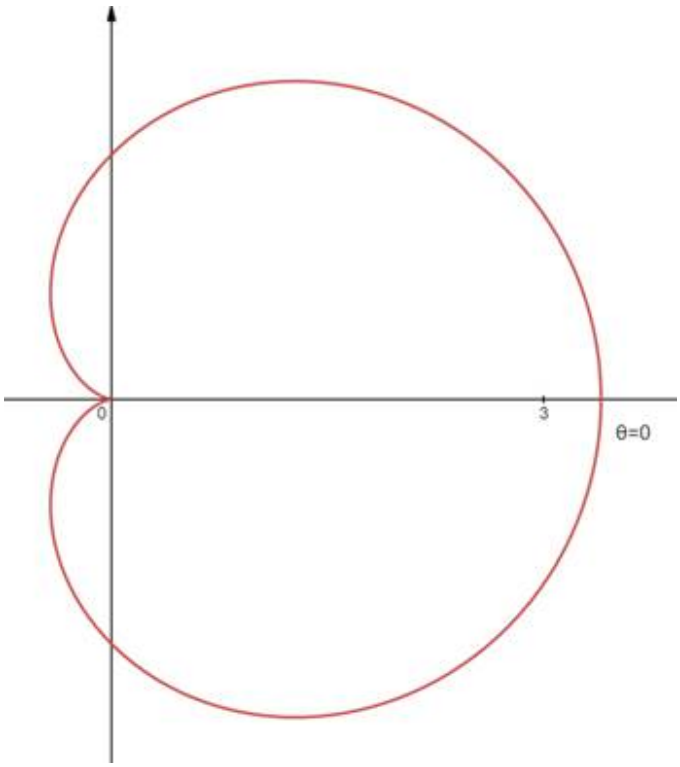
Question	Answer	Marks	Guidar
O.i		B1	Closed c starting a ending at pole, in approxim correct location.
		B1	Cardioid indicatior correct s
		2	
O.ii	$r = a(1+\cos\theta) \Rightarrow \sqrt{x^2+y^2} = a\left(1+\dfrac{x}{\sqrt{x^2+y^2}}\right)$	M1	
	$x^2+y^2 = a(x+\sqrt{(x^2+y^2)})$	A1	Substitut for r **and** cos(θ)
		2	
O.iii	$\text{Sector area} = \dfrac{a^2}{2}\displaystyle\int_0^{\frac{1}{3}\pi}\left(1+2\cos\theta+\cos^2\theta\right)\mathrm{d}\theta$ $= \dfrac{a^2}{2}\displaystyle\int_0^{\frac{1}{3}\pi}\left(\dfrac{3}{2}+2\cos\theta+\dfrac{\cos 2\theta}{2}\right)\mathrm{d}\theta$	M1A1	
	$= \dfrac{a^2}{2}\left[\dfrac{3\theta}{2}+2\sin\theta+\dfrac{\sin 2\theta}{4}\right]_0^{\frac{1}{3}\pi}$	M1	
	$= \dfrac{a^2}{16}\left(4\pi+9\sqrt{3}\right)$	A1	
		4	
O.iv	$\text{Arc length} = \displaystyle\int_0^{\frac{1}{3}\pi}\sqrt{a^2\left(1+2\cos\theta+\cos^2\theta\right)+a^2\left(-\sin\theta\right)^2}\,\mathrm{d}\theta$	M1A1	
	$= a\displaystyle\int_0^{\frac{1}{3}\pi}\sqrt{2+2\cos\theta}\,\mathrm{d}\theta$	A1	

	$= a\left[4\sin\frac{\theta}{2}\right]_0^{\frac{1}{3}\pi} = 2a$	**M1A1**	
		5	

Explanation

O.i

θ	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$
r	2a	$a\left(1+\frac{\sqrt{2}}{2}\right)$	a	$a\left(1-\frac{\sqrt{2}}{2}\right)$	0	$a\left(1-\frac{\sqrt{2}}{2}\right)$	a	$a\left(1+\frac{\sqrt{2}}{2}\right)$



O.ii

use $cos\,\theta = \frac{x}{r}$

$r = a\left(1+\frac{x}{r}\right)$

$r^2 = a(r+x)$

Use $r = \sqrt{x^2+y^2}$ **M1**

$x^2+y^2 = a\left(x+\sqrt{x^2+y^2}\right)$ **A1**

O.iii

Area of sector $=\frac{1}{2}r^2\theta$

Area = $\int_0^{\frac{\pi}{3}} \frac{1}{2} r^2 \, d\theta = \int_0^{\frac{\pi}{3}} \frac{1}{2} (a(1+cos\theta))^2 \, d\theta$

$= \frac{1}{2} \int_0^{\frac{\pi}{3}} a^2 (1 + 2cos\theta + \cos^2\theta) \, d\theta$ **M1A1**

Use double angle formula

$= \frac{a^2}{2} \int_0^{\frac{\pi}{3}} \left(1 + 2cos\theta + \frac{1}{2} + \frac{1}{2} cos2\theta\right) d\theta$

$= \frac{a^2}{2} \left[\frac{3}{2}\theta + 2\sin\theta + \frac{1}{4}\sin 2\theta\right]_0^{\frac{\pi}{3}}$ **M1**

$= \frac{a^2}{2} \left(\frac{3}{2}\frac{\pi}{3} + 2\sin\frac{\pi}{3} + \frac{1}{4}\sin\frac{2\pi}{3} - 0\right)$

$= \frac{\pi a^2}{4} + \frac{9\sqrt{3}a^2}{16}$ **A1**

O.iv

Arc length in polar form is given by: $s = \int \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} \, d\theta$

$\frac{dr}{d\theta} = a(-\sin\theta) = -a\sin\theta$

$s = \int_0^{\frac{\pi}{3}} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} \, d\theta = \int_0^{\frac{\pi}{3}} \sqrt{\left(a(1+cos\theta)\right)^2 + (-a\sin\theta)^2} \, d\theta$ **M1A1**

$= \int_0^{\frac{\pi}{3}} a\sqrt{2 + 2\cos\theta} \, d\theta$ **A1**

$= \int_0^{\frac{\pi}{3}} a\sqrt{2 \cdot 2\cos^2\frac{\theta}{2}} d\theta$ (Use $cos\,2\theta = 2cos^2\theta - 1$)

$= \int_0^{\frac{\pi}{3}} a2\cos\frac{\theta}{2} d\theta$

$= 2a\left[2\sin\frac{\theta}{2}\right]_0^{\frac{\pi}{3}} = 2a\left(2 \cdot \frac{1}{2} - 0\right) = 2a$ **M1A1**

Comment

OR

This was a slightly less popular choice.

i. Sketches were generally clear and most candidates labelled the intercept on the initial line correctly, though some omitted the constant, a. Some cardioids were rotated or translated, but most were correctly positioned.

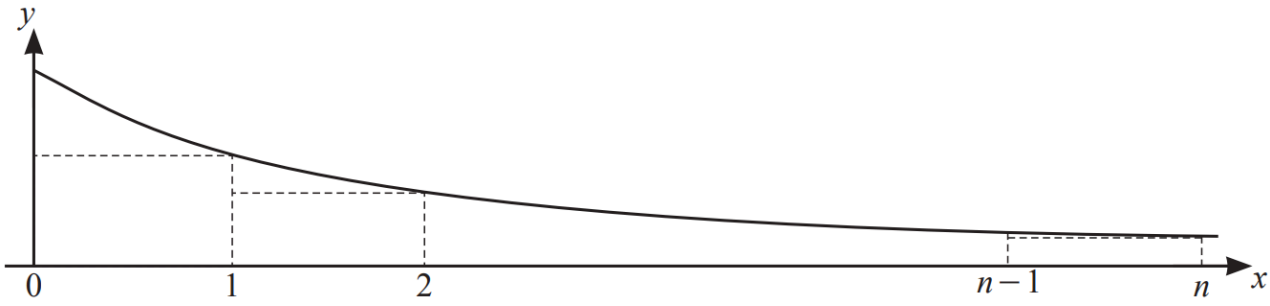
ii. This was well done, as most candidates knew the necessary substitutions, having realised that multiplying both sides by r would allow them to eliminate a and θ from the right hand side.

iii. Those that recalled the formula for sector area correctly found the third part of the question relatively straightforward, and most knew to use the double angle substitution. Few slips were made in the integration then.

iv. This part of the question caused more problems, often because candidates didn't recall the formula needed correctly. Stronger candidates were able to handle the correct integration either by using a trigonometric substitution or by doing integration by substitution.

Answers: **(iii)** $\frac{a^2}{16}\left(4\pi + 9\sqrt{3}\right)$ **(iv)** $2a$

NO. 2



The diagram shows the curve $y = \frac{1}{\sqrt{x^2+x+1}}$ for $x \geq 0$, together with a set of n rectangles of unit width. By considering the sum of the areas of these rectangles, show that

$$\sum_{r=1}^{n} \frac{1}{\sqrt{r^2+r+1}} < \ln\left(\tfrac{1}{3}+\tfrac{2}{3}n+\tfrac{2}{3}\sqrt{n^2+n+1}\right).$$ [10]

A level FURTHER MATHEMATICS 2020/Winter/Paper2_2/Q8

Mark scheme

Question	Answer	Marks	Guidan
8	$\text{Sum} = \frac{1}{\sqrt{3}} + \frac{1}{\sqrt{7}} + \ldots + \frac{1}{\sqrt{\left(n^2+n+1\right)}}$	M1A1	Forms su of the are of the rectangle need to s considera of areas, just the su given in th question.
	$< \int_0^n \frac{1}{\sqrt{\left(x^2+x+1\right)}}\,\mathrm{d}x$	M1	Compare with integ Limits hav to be corr
	$x^2+x+1 = \left(x+\frac{1}{2}\right)^2 + \frac{3}{4}$	B1	Complete the squar
	$\int_0^n \frac{1}{\sqrt{\left(x+\frac{1}{2}\right)^2+\frac{3}{4}}}\,\mathrm{d}x = \left[\sinh^{-1}\left(\frac{x+\frac{1}{2}}{\frac{1}{2}\sqrt{3}}\right)\right]_0^n = \left[\sinh^{-1}\left(\frac{2x+1}{\sqrt{3}}\right)\right]_0^n$	M1 A1	Uses form
	$= \left[\ln\left(\frac{2x+1}{\sqrt{3}} + \sqrt{\frac{(2x+1)^2+3}{3}}\right)\right]_0^n = \left[\ln\left(\frac{2x+1+2\sqrt{\left(x^2+x+1\right)}}{\sqrt{3}}\right)\right]_0^n$	M1 A1	Uses logarithm form of si 1.
	$= \ln\left(\frac{2n+1+2\sqrt{\left(n^2+n+1\right)}}{\sqrt{3}}\right) - \ln\left(\frac{3}{\sqrt{3}}\right)$	M1	Inserts lim
	$= \ln\left(\frac{1}{3} + \frac{2}{3}n + \frac{2}{3}\sqrt{\left(n^2+n+1\right)}\right)$	A1	AG
		10	

Explanation

8.

The exact area is

$A = \int_0^n y\,dx = \int_0^n \frac{1}{\sqrt{x^2+x+1}}dx = \int_0^n \frac{1}{\sqrt{\left(x+\frac{1}{2}\right)^2+\frac{3}{4}}}d\left(x+\frac{1}{2}\right)$ **B1**

(Formula sheet: $\int \frac{1}{\sqrt{x^2+a^2}}dx = sinh^{-1}\left(\frac{x}{a}\right)$)

$= \left[\sinh^{-1}\left(\frac{x+\frac{1}{2}}{\frac{\sqrt{3}}{2}}\right)\right]_0^n = \sinh^{-1}\left(\frac{2n+1}{\sqrt{3}}\right) - \sinh^{-1}\left(\frac{1}{\sqrt{3}}\right)$ **M1 A1**

$= \ln\left(\frac{2n+1}{\sqrt{3}} + \sqrt{\frac{(2n+1)^2}{3}+1}\right) - \ln\left(\frac{1}{\sqrt{3}} + \sqrt{\frac{1}{3}+1}\right)$ **M1 A1**

(Formula sheet: $sinh^{-1} x = ln\left(x + \sqrt{x^2+1}\right)$)

$= \ln\left(\frac{2n+1}{\sqrt{3}} + \sqrt{\frac{4n^2+4n+4}{3}}\right) - \ln\left(\frac{1}{\sqrt{3}} + \frac{2}{\sqrt{3}}\right)$

$= \ln\left(\frac{2n+1}{\sqrt{3}} + \frac{2}{\sqrt{3}}\sqrt{n^2+n+1}\right) - \ln\left(\sqrt{3}\right)$

$= \ln\left(\frac{1}{3} + \frac{2}{3}n + \frac{2}{3}\sqrt{n^2+n+1}\right)$ **M1**

For the lower bound of the area, the *r*th rectangle, the width is 1 ,

the height is $y = \frac{1}{\sqrt{x^2+x+1}} = \frac{1}{\sqrt{r^2+r+1}}$

Sum of areas of rectangles are:

$\sum_{r=1}^{n} 1\left(\frac{1}{\sqrt{r^2+r+1}}\right) = \sum_{r=1}^{n} \frac{1}{\sqrt{r^2+r+1}}$ **M1A1**

So, $\sum_{r=1}^{n} \frac{1}{\sqrt{r^2+r+1}} < \ln\left(\frac{1}{3} + \frac{2}{3}n + \frac{2}{3}\sqrt{n^2+n+1}\right)$ **A1**

Comment

Stronger candidates showed consideration of the sum of the areas of the rectangles to justify the left-hand side of the inequality. When dealing with the right-hand side, the most common approach was to complete the square and use the logarithmic form of the inverse of sin$h$ to justify the given answer. A few candidates worked instead with tan and sec via a suitable substitution and were usually successful too.

NO. 3

a. Sketch the graph of $y = \coth x$ for $x > 0$ and state the equations of the asymptotes. [2]

b. Starting from the definitions of coth and cosech in terms of exponentials, prove that $\coth^2 x - \operatorname{cosech}^2 x = 1$. [3]

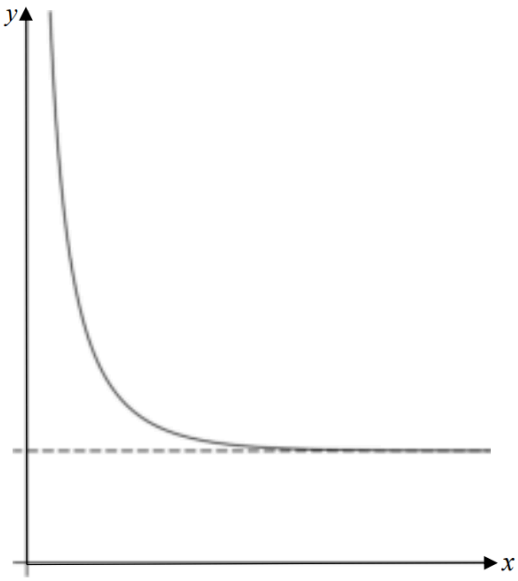
The curve C has equation $y = \ln \coth(\frac{1}{2}x)$ for $x > 0$.

c. Show that $\frac{dy}{dx} = -\operatorname{cosech} x$. [3]

d. It is given that the arc length of C from $x = a$ to $x = 2a$ is $\ln 4$, where a is a positive constant.
Show that $\cosh a = 2$ and find, in logarithmic form, the exact value of a. [7]

A level FURTHER MATHEMATICS 2020/Winter/Paper2_1/Q8

Mark scheme

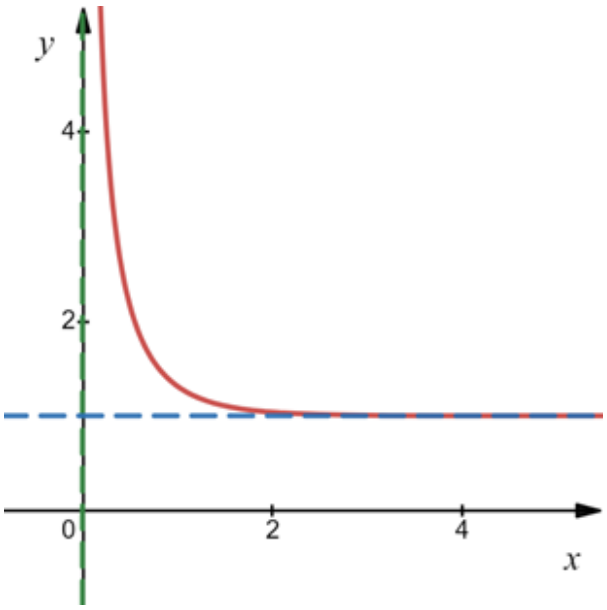
Question	Answer	Marks	Guidance
a		**B1**	Correct shape and position, not too truncated.
	$x = 0$, $y = 1$	**B1**	States equations of asymptotes.
		2	
b	$\coth x = \dfrac{e^x + e^{-x}}{e^x - e^{-x}} \quad \operatorname{cosech} x = \dfrac{2}{e^x - e^{-x}}$	**B1**	
	$\left(\dfrac{e^x + e^{-x}}{e^x - e^{-x}}\right)^2 - \dfrac{4}{\left(e^x - e^{-x}\right)^2} = \dfrac{e^{2x} + e^{-2x} - 2}{\left(e^x - e^{-x}\right)^2} = 1$	**M1** **A1**	Writes over common denominator, AG.
		3	
c	$\dfrac{dy}{dx} = -\dfrac{\operatorname{sech}^2\left(\frac{1}{2}x\right)}{2\tanh\left(\frac{1}{2}x\right)} = -\dfrac{1}{2\sinh\left(\frac{1}{2}x\right)\cosh\left(\frac{1}{2}x\right)}$ Or $\dfrac{dy}{dx} = -\dfrac{\operatorname{cosech}^2\left(\frac{1}{2}x\right)}{2\coth\left(\frac{1}{2}x\right)} = -\dfrac{1}{2\sinh\left(\frac{1}{2}x\right)\cosh\left(\frac{1}{2}x\right)}$	**M1** **A1**	Uses chain rule.
	$= -\dfrac{1}{\sinh(x)} = -\operatorname{cosech} x$	**A1**	AG
		3	
d	$\int_a^{2a} \sqrt{1 + \operatorname{cosech}^2 x}\, dx$	**M1**	Forms correct integral.

	$\int_a^{2a} \sqrt{\coth^2 x}\, \mathrm{d}x = \int_a^{2a} \coth x \, \mathrm{d}x$	**M1 A1**	Uses $\coth^2 x$ – $\text{cosech}^2 x = 1$.
	$= \left[\ln \sinh x\right]_a^{2a} = \ln \sinh 2a - \ln \sinh a$	**M1**	Integrates and substitutes limits.
	$= \ln \dfrac{\sinh 2a}{\sinh a} = \ln\left(2\cosh a\right)$	**M1**	Combines logarithms and uses double angle formula.
	$\ln(2\cosh a) = \ln 4 \Rightarrow \cosh a = 2$	**A1**	AG
	$a = \ln\left(2 + \sqrt{2^2 - 1}\right) = \ln\left(2 + \sqrt{3}\right)$	**A1**	Must reject $\ln\left(2 - \sqrt{3}\right)$.
		7	

Explanation

a.

$y = coth\, x = \frac{e^x + e^{-x}}{e^x - e^{-x}}$



B1(Correct shape and position, not too truncated.)

Vertical asymptote: $e^x - e^{-x} = 0$ →x=0

when x goes to infinity, y goes to 1

Horizontal asymptote: y=1 **B1**

b)

$coth^2 x - cosech^2 x = \left(\frac{e^x + e^{-x}}{e^x - e^{-x}}\right)^2 - \left(\frac{2}{e^x - e^{-x}}\right)^2 = \frac{e^{2x} + e^{-2x} + 2 - 4}{(e^x - e^{-x})^2}$ **B1**

$= \frac{e^{2x} - 2 + e^{-2x}}{(e^x - e^{-x})^2} = \frac{(e^x - e^{-x})^2}{(e^x - e^{-x})^2} = 1$ **M1 A1**

c)

$\frac{dy}{dx} = \frac{d}{dx}\left(\ln\coth\left(\frac{1}{2}x\right)\right) = \frac{1}{\coth\left(\frac{1}{2}x\right)}\frac{d}{dx}\left(\coth\left(\frac{1}{2}x\right)\right) =$

M1 A1

$\frac{1}{\coth\left(\frac{1}{2}x\right)}\left(-\frac{1}{2}\,cosech^2\left(\frac{1}{2}x\right)\right) = -\frac{\sinh\left(\frac{1}{2}x\right)}{2\cosh\left(\frac{1}{2}x\right)\sinh^2\left(\frac{1}{2}x\right)}$

$= -\frac{1}{2\cosh\left(\frac{1}{2}x\right)sinh\left(\frac{1}{2}x\right)} = -\frac{1}{\sinh x} = -\operatorname{cosech} x$ **A1**

$\left(\frac{d}{dx}(\coth x) = \frac{d}{dx}\left(\frac{\cosh x}{\sinh x}\right) = \frac{\sinh^2 x - \cosh^2 x}{\sinh^2 x} = -\frac{1}{\sinh^2 x} = \right.$

$\left. -cosech^2\, x\right)$

d)

Arc length is given by $\int ds = \int \sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx$

$s = \int_a^{2a}\sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx = \int_a^{2a}\sqrt{1+(-cosech\, x)^2}\,dx$ **M1**

$= \int_a^{2a}\sqrt{\coth^2 x}\,dx = \int_a^{2a}\coth x\,dx = [\ln\sinh x]_a^{2a}$ **M1 A1**

$= \ln\sinh 2a - \ln\sinh a = \ln\frac{\sinh 2a}{\sinh a}$ **M1**

So $\ln\frac{\sinh 2a}{\sinh a} = \ln 4$

$sinh\, 2a = 4\, sinh\, a$

$2\, sinh\, a\, cosh\, a = 4\, sinh\, a$ **M1**

$cosh\, a = 2$ **A1**

$a = cosh^{-1} 2 = ln\left(2+\sqrt{2^2-1}\right) = \ln(2+\sqrt{3})$ **A1**

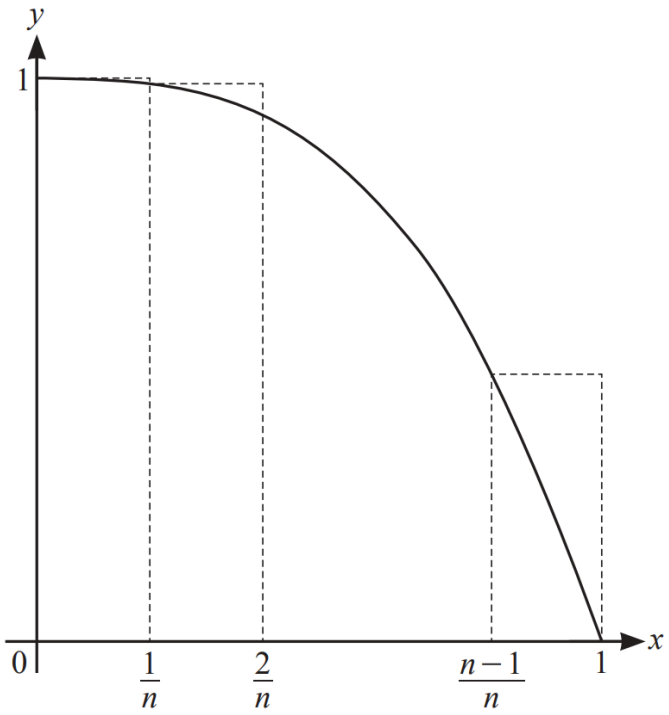
(Use the result in formula sheet)

Comment

a. Most sketches were accurate, showing asymptotes x = 0 and y = 1. A few candidates unnecessarily sketched the curve for x < 0.
b. This part of the question was answered well with the majority of candidates accurately substituting coth and cosech in terms of exponentials and combining using a common denominator.
c. Stronger candidates clearly applied the chain rule and required double angle formula to justify the given derivative.
d. Most candidates accurately recalled the formula for arc length and applied the results given in **(b)** and **(c)** to reach the given answer. Finding the logarithmic form of

$\cosh^{-1}2$ was not problematic for most candidates.

NO. 4



The diagram shows the curve with equation $y = 1 - x^3$ for $0 \leq x \leq 1$, together with a set of n rectangles of width $\frac{1}{n}$.

a. By considering the sum of the areas of the rectangles, show that

$$\int_0^1 (1-x^3)\,dx \leqslant \frac{3n^2+2n-1}{4n^2}.$$ [4]

b. Use a similar method to find, in terms of n, a lower bound for $\int_0^1 (1-x^3)\,dx$. [4]

A level FURTHER MATHEMATICS 2020/Winter/Paper2_1/Q4

Mark scheme

Question	Answer	Marks	Guidan
a	$\int_0^1 1-x^3\,\mathrm{d}x \leqslant \frac{1}{n}+\frac{1}{n}\left(1-\frac{1}{n^3}\right)+\ldots+\frac{1}{n}\left(1-\frac{(n-1)^3}{n^3}\right)$	**M1** **A1**	Forms the sum of of the rectangles.
	$=1-\frac{1}{n^4}\sum_{r=1}^{n-1} r^3 = 1-\frac{(n-1)^2 n^2}{4n^4}$	**M1**	Applies $\sum_{r=1}^{n-1} r^3 = \frac{1}{4}$
	$=\frac{4n^2-(n-1)^2}{4n^2}=\frac{3n^2+2n-1}{4n^2}$	**A1**	AG
		4	
b	$\int_0^1 1-x^3\,\mathrm{d}x \geqslant \frac{1}{n}\left(1-\frac{1}{n^3}\right)+\frac{1}{n}\left(1-\frac{2^3}{n^3}\right)+\ldots+\frac{1}{n}\left(1-\frac{(n-1)^3}{n^3}\right)$	**M1** **A1**	Forms the sum of of appropriate rec
	$=\frac{n-1}{n}-\frac{1}{n^4}\sum_{r=1}^{n-1} r^3 = \frac{n-1}{n}-\frac{(n-1)^2 n^2}{4n^4}$	**M1**	Applies $\sum_{r=1}^{n-1} r^3 = \frac{1}{4}($
	$=\frac{4n(n-1)-(n-1)^2}{4n^2}=\frac{3n^2-2n-1}{4n^2}$	**A1**	
		4	

Explanation

a.

The exact area is

$$A = \int_0^1 y\,dx = \int_0^1 (1-x^3)\,dx$$

For the upper bound of the area, the *r*th rectangle, the width is $\frac{1}{n}$,

the height is $y = 1-x^3 = 1-(\frac{r-1}{n})^3$

Sum of areas of rectangles are:

$$\sum_{r=1}^{n} \frac{1}{n}\left(1-\left(\frac{r-1}{n}\right)^3\right) = \sum_{0}^{n-1} \frac{1}{n}\left(1-\frac{r^3}{n^3}\right) = \sum_{0}^{n-1} \frac{1}{n} - \sum_{0}^{n-1} \frac{r^3}{n^4}$$ **M1 A1**

(Use the standard result $\sum_{r=1}^{n} r^3 = \frac{1}{4}r^2(r+1)^2$)

$$= 1-\frac{1}{n^4}\cdot\left(0+\frac{1}{4}(n-1)^2 n^2\right) = 1-\frac{(n-1)^2}{4n^2} = \frac{4n^2-(n^2-2n+1)}{4n^2}$$ **M1**

$$= \frac{3n^2+2n-1}{4n^2}$$ **A1**

So, $\int_0^1 (1-x^3)\,dx \leq \frac{3n^2+2n-1}{4n^2}$

b)

For the upper of the area, the *r*th rectangle,

the height is $y = 1 - x^3 = 1 - (\frac{r}{n})^3$

$$\sum_{r=1}^{n} \frac{1}{n}\left(1 - \left(\frac{r}{n}\right)^3\right) = \sum_{1}^{n} \frac{1}{n}\left(1 - \frac{r^3}{n^3}\right) = \sum_{1}^{n} \frac{1}{n} - \sum_{1}^{n} \frac{r^3}{n^4}$$ **M1 A1**

$$= 1 - \frac{1}{n^4} \cdot \left(0 + \frac{1}{4}n^2(n+1)^2\right) = 1 - \frac{(n+1)^2}{4n^2} = \frac{4n^2 - (n^2+2n+1)}{4n^2}$$ **M1**

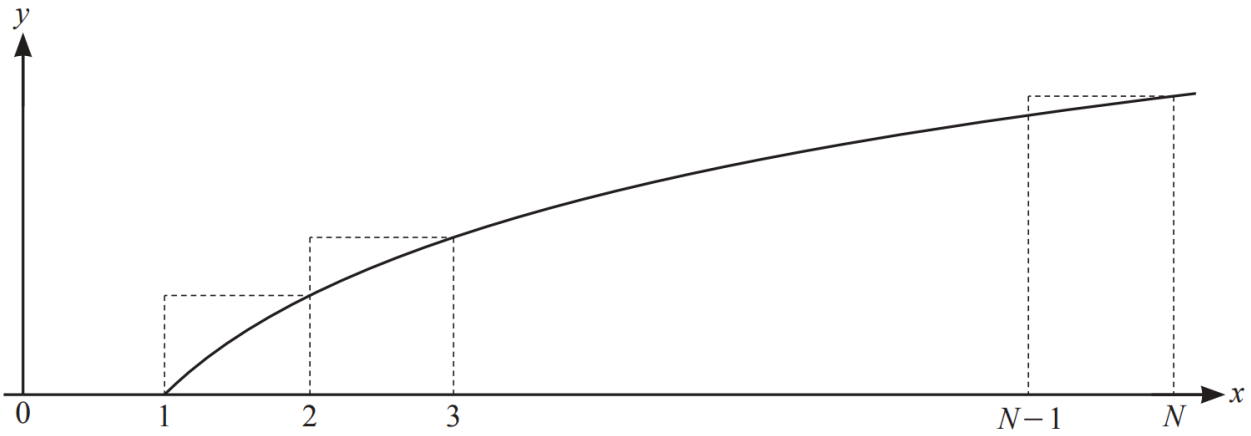
$$= \frac{3n^2 - 2n - 1}{4n^2}$$ **A1**

So $\frac{3n^2 - 2n - 1}{4n^2} \leq \int_0^1 (1 - x^3)\, dx$

Comment

a. Strong candidates formed a correct expression for the sum of the areas of the rectangles and applied the standard result for the sum of cubes to accurately derive the given result.
b. Strong candidates adapted their solution to **(a)** and derived a suitable lower bound.

NO. 5



The diagram shows the curve with equation $y = \ln x$ for $x \geq 1$, together with a set of $(N - 1)$ rectangles of unit width.

a. By considering the sum of the areas of these rectangles, show that $\ln N! > N \ln N - N + 1$. [5]

b. Use a similar method to find, in terms of N, an upper bound for $\ln N!$. [3]

A level FURTHER MATHEMATICS 2020/Summer/Paper2_3/Q4

Mark scheme

Question	Answer	Marks
a	$\ln 2 + \ln 3 + \ldots + \ln N$	**M1**
	$> \int_1^N \ln x \, dx$	**M1**
	$\int_1^N \ln x \, dx = [x \ln x]_1^N - \int_1^N 1 \, dx = N \ln N - N + 1$	**M1** **A1**
	$\ln N! > N \ln N - N + 1$	**A1**
		5
b	$\ln 1 + \ln 2 + \ldots + \ln(N - 1)$ (or $\ln 2 + \ldots + \ln(N - 1)$)	**M1**
	$< \int_1^N \ln x \, dx$ (or $< \int_2^N \ln x \, dx$)	**M1**
	$\ln N! < N \ln N - N + 1 + \ln N = (N + 1)\ln N - N + 1$ (or $\ln N! < (N + 1) \ln N - N + 2(1 - \ln 2)$)	**A1**
		3

Explanation

a.

The exact area is

$$A = \int_1^N y \, dx = \int_1^N \ln x \, dx = [x \ln x - x]_1^N$$

$= (N\ln N - N) - (1\ln 1 - 1) = N\ln N - N + 1$ **M1 A1**

For the upper bound of the exact area, the *r*th rectangle, the width is 1 ,

the height is$y = lnx = \ln(r+1)$ **M1**

Sum of areas of rectangles are:

$\sum_{r=1}^{N-1} \ln(r+1) = \ln 2 + \ln 3 + \ln 4 + \cdots + \ln N = \ln(2 \times 3 \times 4 \times \ldots \times N) = \ln(N!)$ **M1**

So,$\ln(N!) > N\ln N - N + 1$ **A1**

b)

Consider the lower bound of the exact area:

the *r*th rectangle, the width is 1 , the height is$y = lnx = \ln(r)$ **M1**

Sum of areas of rectangles are:

$\sum_{r=1}^{N-1} \ln(r) = \ln 2 + \ln 3 + \ln 4 + \cdots + \ln(N-1) = \ln\big(2 \times 3 \times 4 \times \ldots \times (N-1)\big) = \ln\big((N-1)!\big) = \ln N! - \ln N$ **M1**

So,

$\ln N! - \ln N < N\ln N - N + 1$

$\ln N! < (N+1)\ln N - N + 1$ **A1**

NO. 6

The curves C_1 : y =cosh x and C_2 : y = sinh 2x intersect at the point where $x = a$.

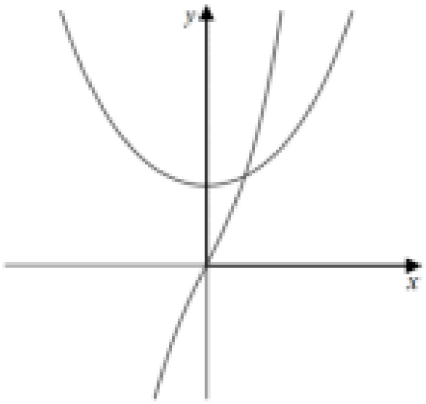
a. Find the exact value of a, giving your answer in logarithmic form. [4]

b. Sketch C_1 and C_2 on the same diagram. [2]

c. Find the exact value of the length of the arc of C_1 from x = 0 to x = a. [5]

A level FURTHER MATHEMATICS 2020/Summer/Paper2_1/Q5

Mark scheme

Question	Answer	Marks
a	cosh a = 2sinh a cosh $a \Rightarrow$ sinh $a = \frac{1}{2}$	**M1** **A1**
	$a = \sinh^{-1}\frac{1}{2} = \ln\left(\frac{1}{2} + \sqrt{\frac{1}{4}+1}\right)$	**M1**
	$a = \ln\left(\frac{1}{2} + \frac{1}{2}\sqrt{5}\right)$	**A1**
		4
b	 (**B1** for C_1 correct, **B1** for C_2 correct and intersecting C_1 in the first quadrant)	**B1** **B1**
		2
c	$\int_0^a \sqrt{1+\sinh^2 x}\,\mathrm{d}x$	**M1**
	$\int_0^a \sqrt{\cosh^2 x}\,\mathrm{d}x = \int_0^a \cosh x\,\mathrm{d}x$	**M1** **A1**
	$\left[\sinh x\right]_0^a = \sinh a$	**M1**
	$\frac{1}{2}$	**A1**
		5

Explanation

a.

solve the equation:$\cosh x = \sinh 2x$

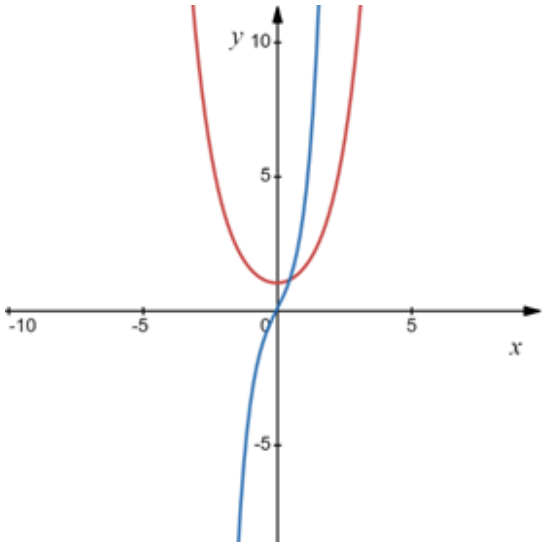
$\cosh x = 2\sinh x \cosh x$

$2\sinh x - 1 = 0$ (always$\cosh x > 0$) **M1 A1**

$a = x = \sinh^{-1}\frac{1}{2} = \ln\left(\frac{1}{2} + \sqrt{\left(\frac{1}{2}\right)^2 + 1}\right) = \ln\left(\frac{1}{2} + \frac{\sqrt{5}}{2}\right)$ **M1A1**

(Use the result of$\sinh^{-1} x$ in formula sheet)

b)



$y = \sinh 2x$ has a similar curve to$y = \sinh x$.

(**B1**forC_1correct,**B1**forC_2correct and intersectingC_1in the first quadrant)

c)

Arc length is given by$\int ds = \int \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$

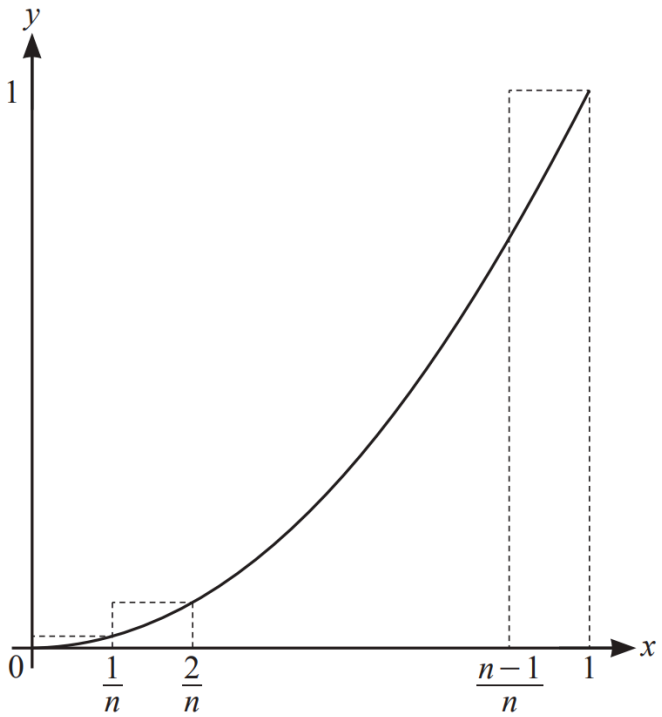
$\frac{dy}{dx} = \sinh x$ **M1**

$$s = \int_0^a \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx = \int_0^a \sqrt{1 + (\sinh x)^2}\, dx = \int_0^a \sqrt{\cosh^2 x}\, dx$$ **M1 A1**

$= \int_0^a \cosh x\, dx = [\sinh x]_0^a = \sinh a = \frac{1}{2}$ **M1A1**

(Use the result in a))

NO. 7



The diagram shows the curve with equation $y = x^2$ for $0 \leq x \leq 1$, together with a set of n rectangles of width $\frac{1}{n}$.

a. By considering the sum of the areas of these rectangles, show that

$$\int_0^1 x^2 \, dx < \frac{2n^2+3n+1}{6n^2}. \quad [4]$$

b. Use a similar method to find, in terms of n, a lower bound for $\int_0^1 x^2 \, dx$. [4]

A level FURTHER MATHEMATICS 2020/Summer/Paper2_1/Q4

Mark scheme

Question	Answer	Marks
a	$\int_0^1 x^2 \, dx < \left(\frac{1}{n}\right)\left(\frac{1}{n}\right)^2 + \left(\frac{1}{n}\right)\left(\frac{2}{n}\right)^2 + \cdots + \left(\frac{1}{n}\right)\left(\frac{n-1}{n}\right)^2 + \left(\frac{1}{n}\right)\left(\frac{n}{n}\right)^2$	**M1 A1**
	$\frac{1}{n^3}\sum_{r=1}^{n} r^2 = \frac{n(n+1)(2n+1)}{6n^3} = \frac{2n^2+3n+1}{6n^2}$	**M1 A1**
		4
b	$\int_0^1 x^2 \, dx > \left(\frac{1}{n}\right)\left(\frac{1}{n}\right)^2 + \left(\frac{1}{n}\right)\left(\frac{2}{n}\right)^2 + \cdots + \left(\frac{1}{n}\right)\left(\frac{n-1}{n}\right)^2$	**M1 A1**
	$= \frac{1}{n^3}\sum_{r=1}^{n-1} r^2 = \frac{(n-1)(n)(2n-2+1)}{6n^3} = \frac{2n^2-3n+1}{6n^2}$	**M1 A1**
		4

Explanation

a.

The exact area is

$$A = \int_0^1 y\,dx = \int_0^1 x^2\,dx$$

For the upper bound of the area, the rth rectangle, the width is $\frac{1}{n}$, the height is $y = x^2 = \left(\frac{r}{n}\right)^2$

Sum of areas of rectangles are:

$$\sum_{r=1}^{n} \frac{1}{n}\left(\frac{r}{n}\right)^2 = \sum_{r=1}^{n} \frac{r^2}{n^3} = \frac{1}{n^3}\frac{1}{6}n(n+1)(2n+1) = \frac{2n^2+3n+1}{6n^2}$$ **M1 A1**

So, $\int_0^1 x^2\,dx < \frac{2n^2+3n+1}{6n^2}$ **M1 A1**

b)

For the lower bound of the area, the rth rectangle, the width is $\frac{1}{n}$, the height is $y = x^2 = \left(\frac{r-1}{n}\right)^2$

Sum of areas of rectangles are:

$$\sum_{r=1}^{n} \frac{1}{n}\left(\frac{r-1}{n}\right)^2 = \sum_{r=0}^{n-1} \frac{1}{n}\left(\frac{r}{n}\right)^2 = 0 + \sum_{r=1}^{n-1} \frac{r^2}{n^3}$$

$$= \frac{1}{n^3}\frac{1}{6}(n-1)n(2n-2+1) = \frac{(n-1)(2n-1)}{6n^2} = \frac{2n^2-3n+1}{6n^2}$$ **M1 A1**

So, $\frac{2n^2-3n+1}{6n^2} < \int_0^1 x^2\,dx$ **M1 A1**

NO. 8

The curve C has parametric equations

$$x = \frac{1}{2}e^{2t} - \frac{1}{3}t^3 - \frac{1}{2}, \qquad y = 2e^t(t-1), \qquad \text{for } 0 \leqslant t \leqslant 1.$$

Find the exact length of C. [7]

A level FURTHER MATHEMATICS 2024/Winter/Paper2_2/Q3

Mark scheme

Question	Answer	Marks	Guidance
3	$\dot{x} = e^{2t} - t^2,\ \dot{y} = 2e^t + 2(t-1)e^t = 2te^t$	B1B1	Differentiates x and y with respect to t.
	$\dot{x}^2 + \dot{y}^2 = \left(e^{2t} - t^2\right)^2 + 4t^2e^{2t} = e^{4t} + 2t^2e^{2t} + t^4 = \left(e^{2t} + t^2\right)^2$	M1A1	Factorises $\dot{x}^2 + \dot{y}^2$.
	$\int_0^1 e^{2t} + t^2 \, dt = \left[\frac{1}{2}e^{2t} + \frac{1}{3}t^3\right]_0^1$	M1A1	Applies correct formula for arc length.
	$= \frac{1}{2}e^2 - \frac{1}{6}$	A1	
		7	

Comment

The majority of candidates accurately recalled the formula for the length of the curve with correct limits. Most candidates fully simplified $\sqrt{\dot{x}^2 + \dot{y}^2}$ before substituting into the formula, which caused fewer errors and enabled a clear path to the answer.

NO. 9

A curve has equation $y = e^x$ for $\ln\frac{4}{3} \leqslant x \leqslant \ln\frac{12}{5}$. The area of the surface generated when the curve is rotated through 2π radians about the x-axis is denoted by A.

a. Use the substitution $u = e^x$ to show that $A = 2\pi\int_{\frac{4}{3}}^{\frac{12}{5}} \sqrt{1+u^2}\,\mathrm{d}u$. [2]

b. Use the substitution $u = \sinh v$ to show that $A = \pi\left(\frac{904}{225} + \ln\frac{5}{3}\right)$. [6]

A level FURTHER MATHEMATICS 2024/Winter/Paper2_1/Q3

Mark scheme

Question	Answer	Marks	Guidance
a	$A = 2\pi\int_{\ln\frac{4}{3}}^{\ln\frac{12}{5}} e^x\sqrt{1+e^{2x}}\,\mathrm{d}x = 2\pi\int_{\frac{4}{3}}^{\frac{12}{5}} u\sqrt{1+u^2}\left(\frac{1}{u}\right)\mathrm{d}u$	M1A1	Correct formula and applies substitution, AG.
		2	
b	$u = \sinh v \Rightarrow \mathrm{d}u = \cosh v\,\mathrm{d}v$	B1	
	$A = 2\pi\int_{\ln 3}^{\ln 5} \sqrt{1+\sinh^2 v}\cosh v\,\mathrm{d}v \quad A = 2\pi\int_{\ln 3}^{\ln 5} \cosh^2 v\,\mathrm{d}v$	M1A1	Applies substitution and $\cosh^2 x = 1+\sinh^2 x$. Need limits for A1.
	$\pi\int_{\ln 3}^{\ln 5} (\cosh 2v + 1)\,\mathrm{d}v$	M1	Applies $2\cosh^2 x = \cosh 2x + 1$.
	$= \pi\left[\frac{1}{2}\sinh 2v + v\right]_{\ln 3}^{\ln 5}$	A1	Correct integration.
	$\pi\left(\frac{312}{50} + \ln 5 - \frac{40}{18} - \ln 3\right) = \pi\left(\frac{904}{225} + \ln\frac{5}{3}\right)$	A1	AG.
		6	

Comment

a. Most candidates accurately recalled the formula for the surface area generated by a curve and showed sufficient working when applying the substitution to justify the given answer.
b. The majority of candidates used the given substitution and the required hyperbolic identity to simplify the integral to $\int\cosh^2 v\,dv$, which enabled a clear path to the

answer. Good responses showed the change of limits clearly using the logarithmic form of $\sinh^{-1} u$.